AML vs ALL Classification Using Machine Learning and Graph Neural Networks

Abrar Ahmad, Mehar Ali, Wahaj Aziz, Hamza Farooq
Department of Computer Science
Information Technology University

***Abstract*—We present a machine-learning pipeline to distinguish Acute Myeloid Leukemia (AML) from Acute Lymphoblastic Leukemia (ALL) using microscopic blood-smear images. The dataset (IML Lab, ITU) contains images captured at multiple magnifications and camera setups. We convert detected lymphocyte bounding boxes into graph representations using spatial distances as edge weights and evaluate several Graph Neural Network (GNN) architectures (GIN variants, GraphSAGE, GAT) and a Graph Transformer. Results show GIN variants achieve the best performance on this dataset. The paper includes methodology, model descriptions, comparative results, and discussion of limitations and future directions.**

***Index Terms*—Leukemia, AML, ALL, Graph Neural Networks, Medical Imaging, Machine Learning**

I. Introduction

Leukemia is a malignancy of blood-forming tissues characterized by uncontrolled white blood cell proliferation. Accurate subtyping (e.g., AML vs ALL) is crucial for appropriate treatment. Manual microscopy is time-consuming and subject to inter-observer variability. Automated image-based classification using machine learning can assist pathologists by providing fast, repeatable predictions.

This work focuses on converting per-cell detections into graphs that capture spatial relations between cells and training GNNs to classify the whole-image graph as AML or ALL.

II. Prior Work

Our work is built upon the following research: The original study collected blood samples from patients clinically diagnosed with leukemia and processed them under controlled laboratory conditions. Experts manually annotated each image by identifying individual lymphocytes and drawing bounding boxes around them, ensuring that every labeled cell was correctly associated with its underlying clinical subtype—either Acute Lymphoblastic Leukemia (ALL) or Acute Myeloid Leukemia (AML) or any other types. This expert-verified labeling process served as the foundation for generating high-quality ground-truth datasets, enabling downstream machine learning models to learn morphological distinctions between AML and ALL at the cellular level. The precise, expert-guided bounding-box annotations ensured reliability, minimized label noise, and provided a robust basis for graph-based representation and classification.

III. Our Contribution

We build a classification pipeline that:

- Cleans and prepares the AML/ALL subset of the dataset (removed other types like CML, e.t.c).
- Converts YOLO-style bounding boxes of lymphocytes into graph structures.
- Trains and compare multiple model architectures.
- Evaluates models using accuracy, F1-score, and confusion matrices.

IV. Dataset (Source and Explanation)

A. Source

The dataset used is the Blood Cancer Dataset (Leukemia) released by Intelligent Machines Lab, ITU (MICCAI 2024). The repository URL is provided in the project materials.

B. Description

Images were captured under three magnifications (100X, 40X, 10X) and three setups (high-cost microscope camera, low-cost microscope camera, and mobile phone — Honor 9X Lite). Naming scheme: `H/L_100X/40X/10X_C1/C2`, where C1 denotes mobile camera and C2 denotes microscope camera (H/L distinguishes camera quality).

Most images belong to AML and ALL; minority classes (e.g., CML) were too few and excluded to avoid noisy labels.

V. Methodology

Our methodology contains four steps: cleaning, graph construction, model training, and evaluation.

A. Data cleaning and splitting

We filtered the repository to keep only AML and ALL images, and we didn't mix the original train and test set due to similarities between their test and train images.

B. Graph construction

For each image, detected cell bounding boxes are converted into nodes. Weights are added between nodes as the inverse distance between nodes.

C. Model training

We trained graph-level classifiers that take the image-graph as input and output a binary label (AML/ALL). Models trained include:

- **GIN**: 2-layer, 3-layer, JK, Virtual Node and pool variants tested.
- **GraphSAGE**: 2-layer, 3-layer and JK variants
- **GAT**: 2-layer and 3-layer variants
- **Training Details:**
  - Loss: `CrossEntropyLoss`
  - Optimizer: Adam
  - Learning rate scheduler: `ReduceLROnPlateau` (monitor validation accuracy)
  - Early stopping enabled

D. Evaluation

We evaluated models on held-out test data using accuracy, precision, recall, F1-score, and confusion matrices.

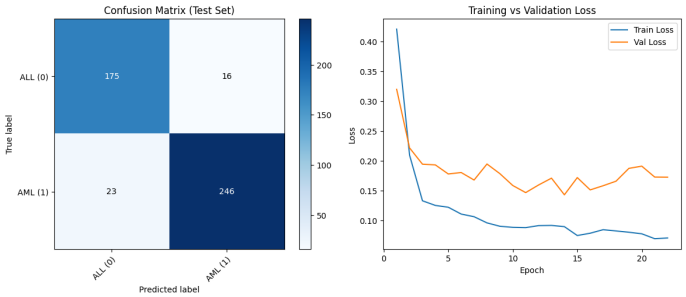
VI. Models (Details and Results)

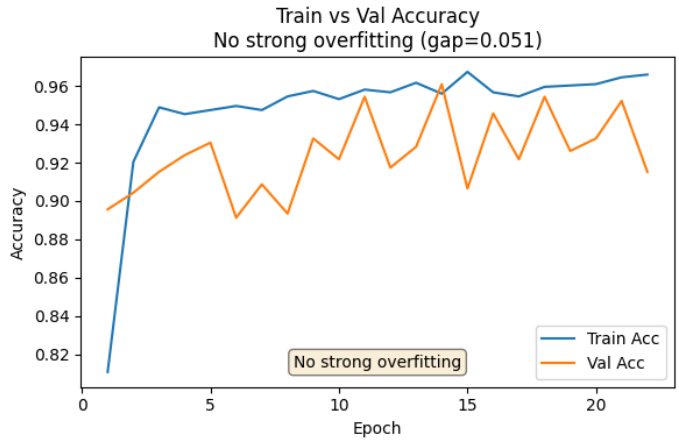
Below we summarize each model family:

A. Graph Isomorphism Network (GIN)

1) GIN:

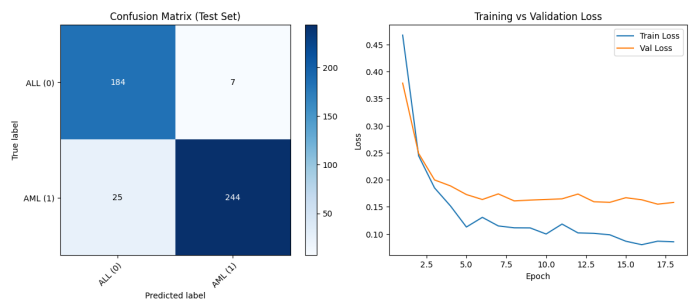
- 3 layers of **GINConv**
  - Each layer's MLP: Linear $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Linear $\rightarrow$ ReLU
- **Readout:**
  - Layer-wise pooled embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$, $g_3 = global_add_pool(h_3)$
  - Final representation: concat$([g_1, g_2, g_3]) \rightarrow$ MLP $\rightarrow$ 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9659
  - Best Validation Accuracy: 0.9609
  - Validation Accuracy (final): 0.9152
  - Test Accuracy: 0.9152

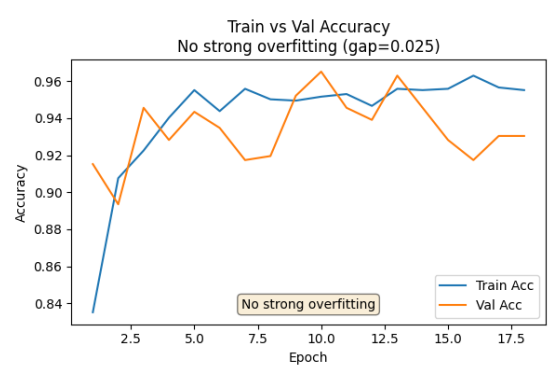




2) GIN (2 Layers):

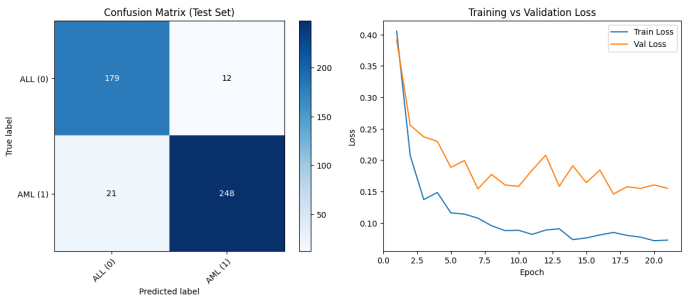
- 2 layers of **GINConv**
- **Readout:**
  - Layer-wise pooled embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$
  - Final representation: concat$([g_1, g_2]) \rightarrow$ MLP $\rightarrow$ 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9552
  - Best Validation Accuracy: 0.9652
  - Validation Accuracy (final): 0.9304
  - Test Accuracy: 0.9304

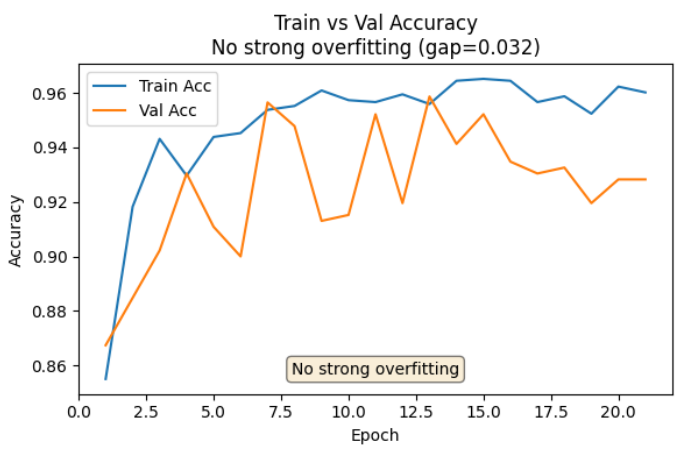




3) GIN with Virtual Node:

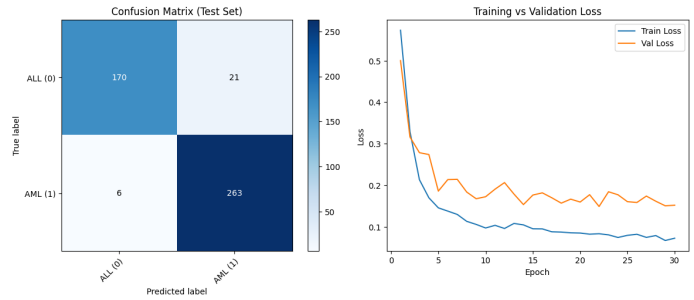
- 3 layers of **GINConv** (each with internal MLP)
- **Virtual Node Embedding:**
  - 1 learnable virtual node vector per graph
  - Node embeddings updated each layer with virtual node
  - Virtual node updated using pooled node features (`global_add_pool`) + MLP
  - Layer-wise pooled embeddings combined → classifier → 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9652
  - Best Validation Accuracy: 0.9283
  - Validation Accuracy (final): 0.9065
  - Test Accuracy: 0.9065

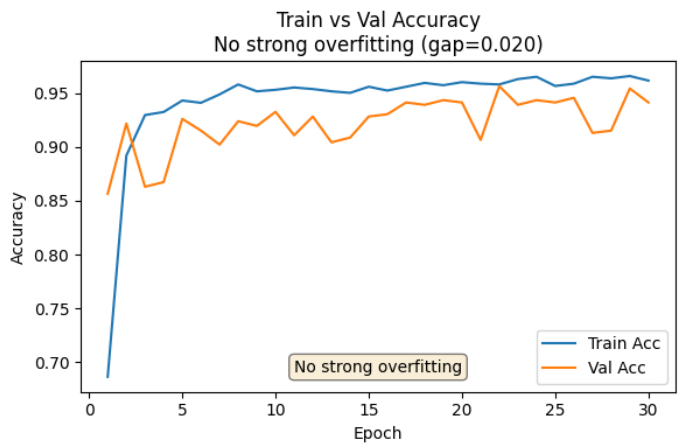




4) GIN with JK:

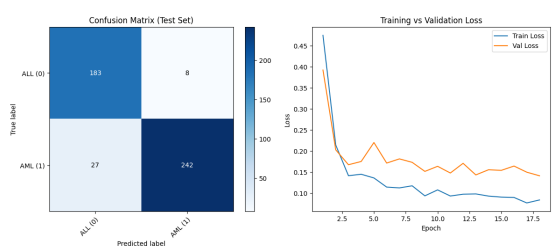
- 3 layers of **GINConv** (each with internal MLP)
- **Jumping Knowledge Readout:**
  - Layer-wise pooled embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$, $g_3 = global_add_pool(h_3)$
  - Stack embeddings: $g_stack = \text{stack}([g_1, g_2, g_3])$
  - Final representation: $\max(g_stack \text{ over layers})$ → classifier → 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9744
  - Best Validation Accuracy: 0.9413
  - Validation Accuracy (final): 0.9413
  - Test Accuracy: 0.9413

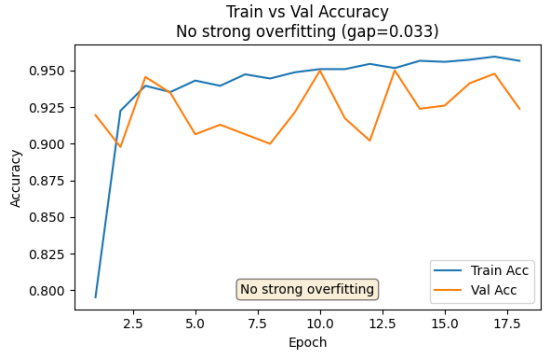




5) GIN with Pooling:

- 3 layers of **GINConv** (each with internal MLP)
- **Pooling Readout:**
  - Uses `GlobalAttention` pooling as attention-based readout
  - Also uses sum pooling (`global_add_pool`); combined readout → classifier → 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9659
  - Best Validation Accuracy: 0.9435
  - Validation Accuracy (final): 0.9217
  - Test Accuracy: 0.9217

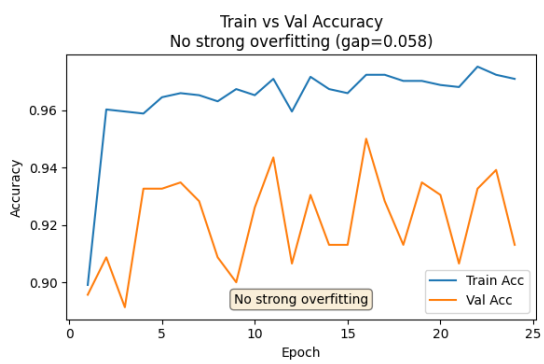


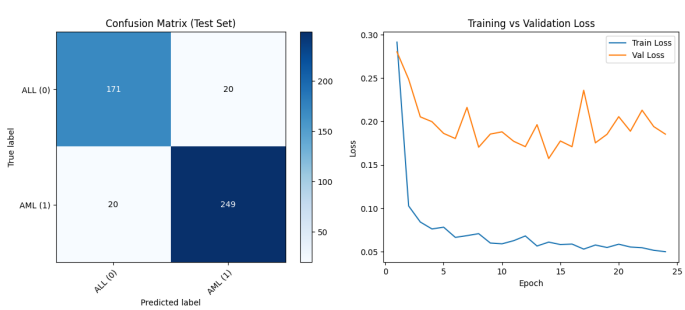


B. GraphSAGE

1) GraphSAGE:

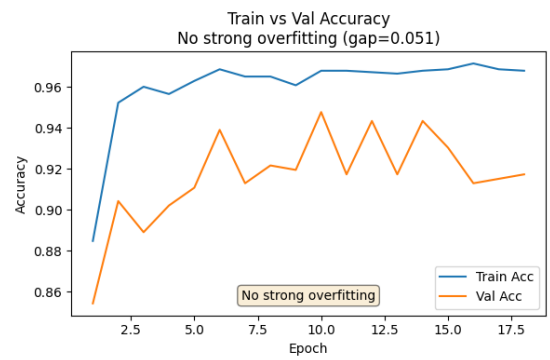
- 3 layers of **SAGEConv**
- **Readout:**
  - Layer-wise pooled embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$, $g_3 = global_add_pool(h_3)$
  - Final representation: $\text{concat}([g_1, g_2, g_3]) \rightarrow$ MLP $\rightarrow$ 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9531
  - Best Validation Accuracy: 0.9391
  - Validation Accuracy (final): 0.9130
  - Test Accuracy: 0.9130

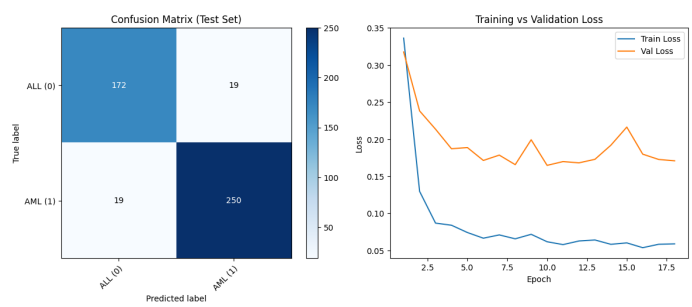




2) GraphSAGE (2 Layers):

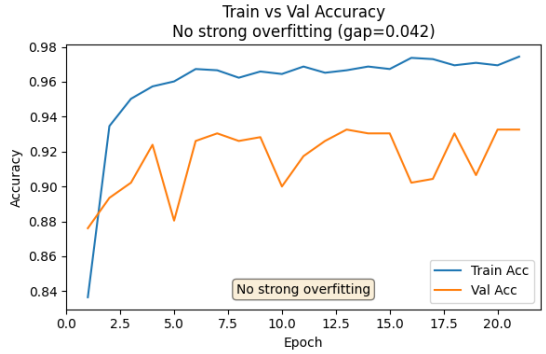
- 2 layers of **SAGEConv**
- **Readout:**
  - Layer-wise pooled embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$
  - Final representation: $\text{concat}([g_1, g_2]) \rightarrow$ MLP $\rightarrow$ 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9680
  - Best Validation Accuracy: 0.9478
  - Validation Accuracy (final): 0.9174
  - Test Accuracy: 0.9174

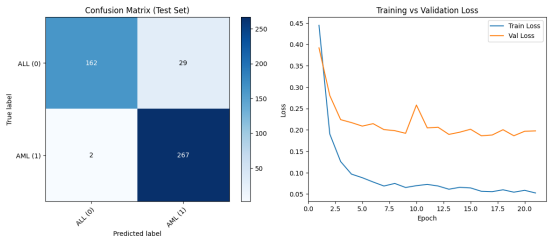




3) GraphSAGE with JK:

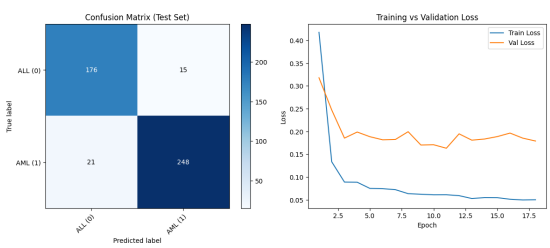
- 3 layers of **SAGEConv**
- **JK-style Readout:**
  - Layer-wise graph embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$, $g_3 = global_add_pool(h_3)$
  - Jumping Knowledge combination: $g_{stack} = stack([g_1, g_2, g_3])$, $h = \max(g_{stack}$ over layers$)$
  - Final MLP $\rightarrow$ 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9744
  - Best Validation Accuracy: 0.9326
  - Validation Accuracy (final): 0.9326
  - Test Accuracy: 0.9326

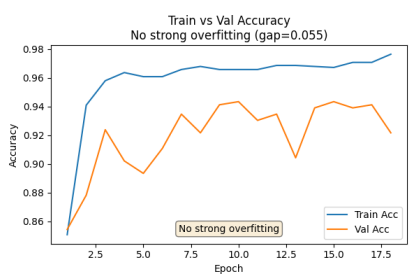




4) GraphSAGE with Pooling:

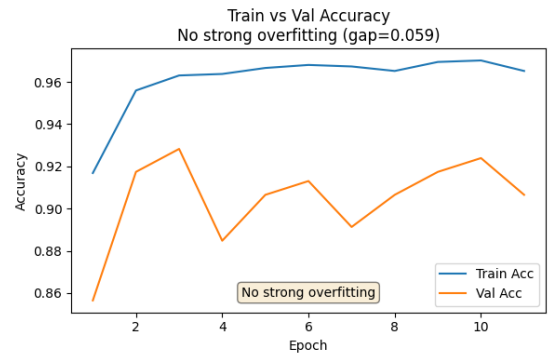
- 3 layers of **SAGEConv**
- **Pooling / Readout:**
  - Attention-based readout: `GlobalAttention` pooling
  - Additional sum pooling: `global_add_pool`
  - Combined pooled embeddings passed to classifier head → 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9765
  - Best Validation Accuracy: 0.9435
  - Validation Accuracy (final): 0.9217
  - Test Accuracy: 0.9217

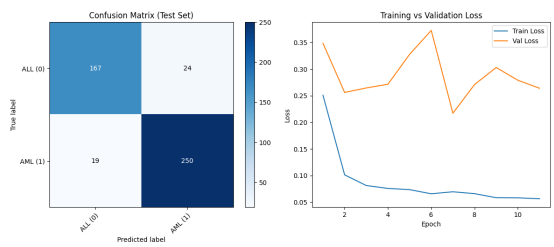




5) GraphSAGE with Virtual Node:

- 3 layers of **SAGEConv**
- **Virtual Node Embedding:**
  - 1 learnable virtual node vector per graph
  - Node embeddings updated each layer with virtual node
  - Virtual node updated using pooled node features (`global_add_pool`) + MLP
  - Layer-wise pooled embeddings concatenated/aggregated → classifier → 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9652
  - Best Validation Accuracy: 0.9283
  - Validation Accuracy (final): 0.9065
  - Test Accuracy: 0.9065

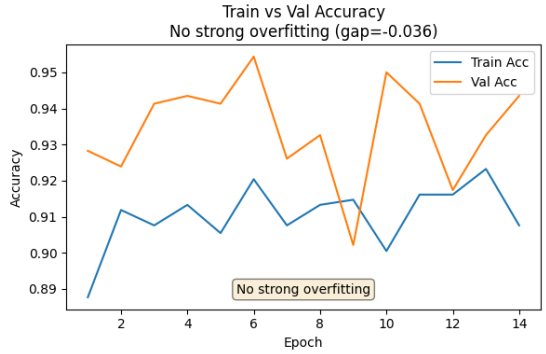


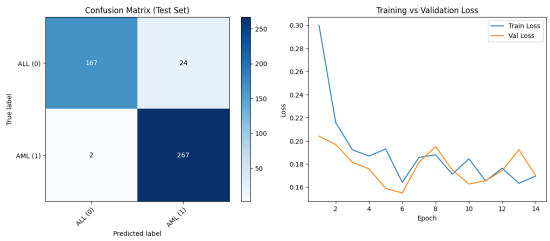


C. Graph Attention Network (GAT)

1) GAT:

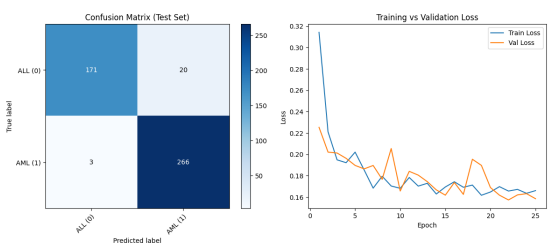
- 3 layers of **GATConv**
- **Readout:**
  - Layer-wise pooled embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$, $g_3 = global_add_pool(h_3)$
  - Final representation: concat$([g_1, g_2, g_3])$ → MLP → 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9126
  - Best Validation Accuracy: 0.9587
  - Validation Accuracy (final): 0.9500
  - Test Accuracy: 0.9500

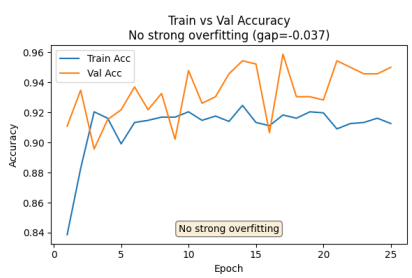




2) GAT (2 Layers):

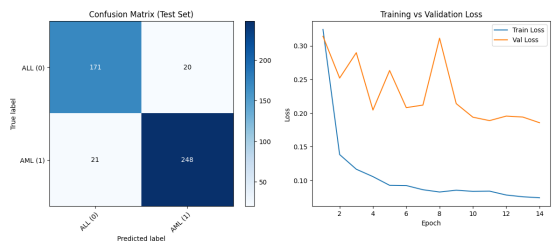
- 2 layers of **GATConv**
- **Readout:**
  - Layer-wise pooled embeddings: $g_1 = global_add_pool(h_1)$, $g_2 = global_add_pool(h_2)$
  - Final representation: concat$([g_1, g_2]) \rightarrow$ MLP $\rightarrow$ 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9552
  - Best Validation Accuracy: 0.9652
  - Validation Accuracy (final): 0.9304
  - Test Accuracy: 0.9304

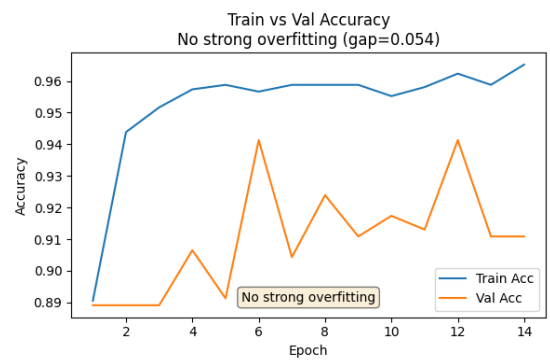




3) GAT with Virtual Node:

- 3 layers of **GATConv**:
  - Hidden dimension: 64
  - Attention heads: 4
  - `concat=False`
- **Virtual Node embedding:**
  - 1 learnable vector per graph
  - Updated after each layer using pooled node features (`global_add_pool`) + MLP
- **Readout:**
  - Pooled embeddings from all 3 layers (g_1, g_2, g_3) concatenated
  - Passed through an MLP to produce 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9652
  - Best Validation: 0.9413
  - Test Accuracy: 0.9109

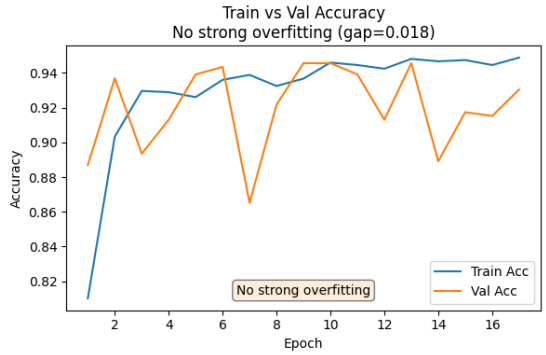


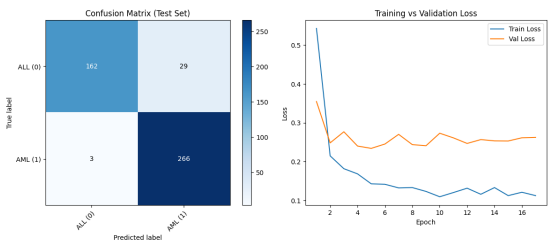


4) GAT with Pooling:

- 3 layers of **GATConv**:
  - Hidden dimension: 64
  - Attention heads: 4
  - `concat=False`
- **Pooling / Readout:**
  - Attention-based readout: `GlobalAttention` pooling
  - Additional `global_add_pool` (sum pooling)
  - Pooled embeddings passed to classifier head for 2 logits (AML vs ALL)
- **Performance:**
  - Train Accuracy: 0.9488

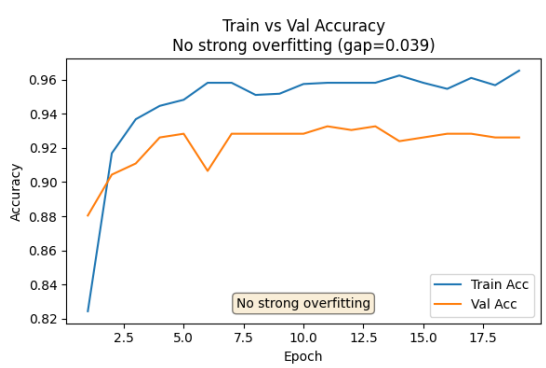
- ∗ Best Validation Accuracy: 0.9457
- ∗ Validation Accuracy (final): 0.9304
- ∗ Test Accuracy: 0.9304

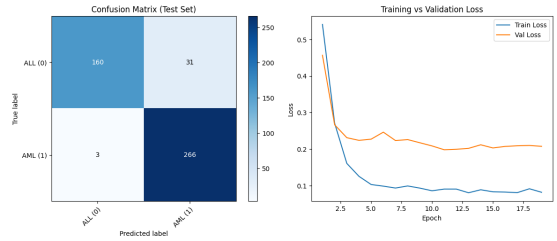




5) *GAT with JK:*

- 3 layers of **GATConv**:
  - ∗ Hidden dimension: 64
  - ∗ Attention heads: 4
  - ∗ `concat=False`
- **JK-style Readout:**
  - ∗ Pooled embeddings from each layer using `global_add_pool`: $g_1 = pool(h_1)$, $g_2 = pool(h_2)$, $g_3 = pool(h_3)$
  - ∗ Final representation: concat$([g_1, g_2, g_3]) \rightarrow$ MLP
  - ∗ Outputs 2 logits (AML vs ALL)
- **Performance:**
  - ∗ Train Accuracy: 0.9652
  - ∗ Best Validation Accuracy: 0.9326
  - ∗ Validation Accuracy (final): 0.9261
  - ∗ Test Accuracy: 0.9261

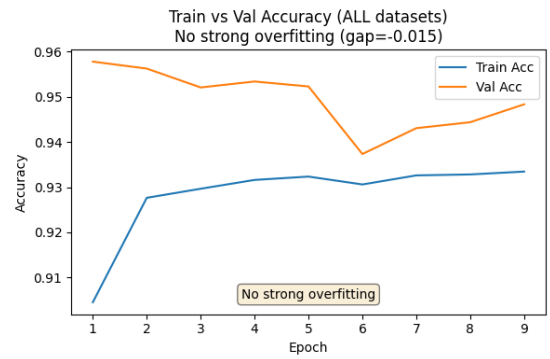


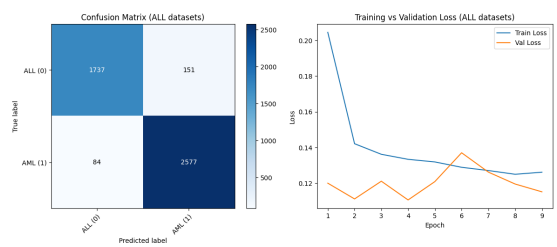


VII. Best Performer Stats on all datasets

The best performer was GraphSage with Jk with following stats:

- **Performance:**
  - Train Accuracy: 0.9744
  - Best Validation Accuracy: 0.9326
  - Validation Accuracy (final): 0.9261
  - Test Accuracy: 0.9261





VIII. Challenges and Limitations

- **Unavailability of Healthy Dataset:**
- **Class imbalance:** Minority leukemia types have few samples.
- **Clinical validation:** Further validation with pathologists and external datasets is required.

IX. Conclusion

We demonstrate that graph-based representations of detected blood cells combined with GNNs are effective for AML vs ALL classification. Graph Sage JK variant produced the strongest results in our experiments.

X. Future Work

We collect healthy samples under the same microscope and imaging conditions, enabling consistent data for future expansion toward distinguishing Healthy vs. Ill patients using our AML–ALL analysis framework.

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References